VB.NET Slot Machine – Student Instruction Manual

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You will build a slot machine in Visual Basic (Windows Forms). Three reels show pictures chosen by random numbers, you bet from a balance, matches pay your bet times a multiplier, and your last 10 spins are kept in an array.

Required features

Your program must have all four features below. Your teacher will run your program and check each one.

Feature	What it must do	How it will be checked
Slots with images	Three reels. Each spin uses random numbers to choose a symbol for each reel, and each reel shows that symbol's picture.	Spin several times: all three reels change to pictures, not text, and the results are different each time.
Balance	Starts at $100. The bet is taken away on every spin and winnings are added. When the balance is $0 you cannot spin anymore.	The balance label updates after every spin. At $0 the Spin button is disabled.
Winnings with multipliers	Three of a kind pays bet × the symbol's multiplier. A pair pays bet × 1. No match pays nothing.	The result label shows the bet and the amount won, and the balance matches.
Spin history	Every spin is stored in an array that holds a maximum of 10 spins. When it is full the oldest spin is removed.	After 11 or more spins the history list shows exactly 10 entries, newest at the top.

Payout table (use these values):

Result	Multiplier
3 × Cherry	×2
3 × Lemon	×3
3 × Bell	×5

Result	Multiplier
3 × Star	×10
3 × Seven	×25
Any pair	×1 (bet returned)
No match	×0 (bet lost)

Before you start

You need Visual Studio and five pictures.

- **Visual Studio 2022** (Community edition is free) with the **.NET desktop development** workload installed.
- **Five pictures**, one for each symbol: cherry, lemon, bell, star and seven. PNG files with a transparent background look best. Any size works; they are scaled to fit.
- Rename the picture files exactly like this, with a capital first letter: `Cherry.png`, `Lemon.png`, `Bell.png`, `Star.png`, `Seven.png`. You may use `.jpg`, `.gif` or `.bmp` instead of `.png`.

Step 1: Create the project

1. Open Visual Studio and click **Create a new project**.
2. In the search box type `Windows Forms`, then choose **Windows Forms App** with the **Visual Basic** tag. Do not choose the C# one.
3. Click **Next**. Name the project `SlotMachine` and choose where to save it.
4. Click **Next**, keep the default framework, and click **Create**.

Visual Studio opens an empty form called `Form1`. You will put all your controls on this form.

Step 2: Build the form

Drag each control below from the **Toolbox** onto `Form1`, then set its **(Name)** in the **Properties** window (press F4 if you cannot see it). The names must match exactly, because your code uses them.

Control	(Name)	Settings
PictureBox	`picReel1`	Size 130 × 120, `SizeMode` = Zoom, `BorderStyle` = FixedSingle, `BackColor` = White
PictureBox	`picReel2`	Same as `picReel1`, placed to its right
PictureBox	`picReel3`	Same as `picReel1`, placed to the right of `picReel2`
Label	`lblBalance`	`Text` = Balance: $100, bold font
Label	`lblBetCaption`	`Text` = Bet: $ (just a caption)
TextBox	`txtBet`	Where the player types the bet
Button	`btnSpin`	`Text` = SPIN
Label	`lblResult`	Tall enough for 2 lines of text
Label	`lblHistoryCaption`	`Text` = Last 10 spins (newest first):
ListBox	`lstHistory`	About 430 × 160, wide enough for one spin per line

Suggested layout, top to bottom:

1. A title label, for example VB SLOTS (optional).
2. The three reel PictureBoxes side by side.
3. `lblBalance`.
4. `lblBetCaption`, `txtBet` and `btnSpin` on one row.
5. `lblResult`.
6. `lblHistoryCaption`, then `lstHistory`.

Tip: click the form itself and set its `AcceptButton` property to `btnSpin`. Then pressing Enter also spins.

Step 3: Add the pictures

The program loads the pictures from a folder called `Images` that is copied next to your program when it runs.

1. In **Solution Explorer**, right-click your **project name** (the line under "Solution") and choose **Add → New Folder**. Name it `Images`.
2. Drag your five picture files from File Explorer onto the `Images` folder in Solution Explorer.

3. Select all five pictures: click the first, then hold **Ctrl** and click the others.
4. In the **Properties** window set:
   - **Build Action** → `Content`
   - **Copy to Output Directory** → `Copy if newer`

Do not skip step 4. Without it the pictures are not copied and the reels will show text instead of images.

Step 4: Variables and arrays

Right-click `Form1` in Solution Explorer and choose **View Code**. Type the following inside `Public Class Form1`, at the top, before any `Sub`.

```
' Symbols that can appear on a reel
Private symbols() As String = {"Cherry", "Lemon", "Bell", "Star", "Seven"}

' Multiplier for three of a kind (same index as symbols)
Private multipliers() As Integer = {2, 3, 5, 10, 25}

' Multiplier for any two matching symbols (bet is returned)
Private Const PAIR_MULTIPLIER As Integer = 1

' History of spins - holds a maximum of 10 spins
Private Const MAX_HISTORY As Integer = 10
Private history(MAX_HISTORY - 1) As String
Private historyCount As Integer = 0

' Picture for each symbol (same index as symbols)
Private symbolImages(4) As Image

Private Const STARTING_BALANCE As Integer = 100
Private balance As Integer = STARTING_BALANCE
Private rnd As New Random()
```

What each part does:

- `symbols`, `multipliers` and `symbolImages` are **parallel arrays**: index 0 is Cherry, its multiplier 2 and its picture; index 4 is Seven, ×25 and its picture.
- `history(MAX_HISTORY - 1)` creates an array with indexes 0 to 9, so it holds exactly 10 spins. `historyCount` tracks how many slots are used.
- `balance` is the player's money. `rnd` is the random number generator used for the reels.

Step 5: Load the pictures when the form opens

`Form1_Load` runs once when the program starts. It loads the five pictures into `symbolImages`, clears the reels and shows the starting balance. Add these below your variables.

```
Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load
    LoadSymbolImages()
    picReel1.Image = Nothing
    picReel2.Image = Nothing
    picReel3.Image = Nothing
    lblResult.Text = "Enter a bet and press Spin!"
    txtBet.Text = "10"
    UpdateBalance()
End Sub

Private Sub LoadSymbolImages()
    Dim missing As String = ""

    ' Pictures are in an "Images" folder next to the program's .exe
    Dim folder As String = IO.Path.Combine(Application.StartupPath, "Images")
    Dim extensions() As String = {".png", ".jpg", ".jpeg", ".gif", ".bmp"}

    For i As Integer = 0 To symbols.Length - 1
        Dim img As Image = Nothing

        ' Look for a file named after the symbol, e.g. Images\Cherry.png
        For Each ext As String In extensions
            Dim path As String = IO.Path.Combine(folder, symbols(i) & ext)
            If IO.File.Exists(path) Then
                img = Image.FromFile(path)
                Exit For
            End If
        Next

        If img Is Nothing Then
            ' No picture found: draw the symbol's name instead
            missing &= symbols(i) & " "
            img = MakeTextImage(symbols(i))
        End If

        symbolImages(i) = img
    Next

    If missing <> "" Then
        MessageBox.Show("No picture found in " & folder & " for: " & missing
```

```
& Environment.NewLine &
                        "The name will be shown instead.", "Missing
pictures")
    End If
End Sub

Private Function MakeTextImage(text As String) As Image
    ' Creates a simple picture with the symbol's name written on it
    Dim bmp As New Bitmap(130, 120)
    Using g As Graphics = Graphics.FromImage(bmp)
        g.Clear(Color.White)
        Using f As New Font("Segoe UI", 16, FontStyle.Bold)
            Dim fmt As New StringFormat()
            fmt.Alignment = StringAlignment.Center
            fmt.LineAlignment = StringAlignment.Center
            g.DrawString(text, f, Brushes.Black, New RectangleF(0, 0, 130,
120), fmt)
        End Using
    End Using
    Return bmp
End Function

Private Sub UpdateBalance()
    lblBalance.Text = "Balance: $" & balance
    ' Disable the Spin button when there is no money left
    btnSpin.Enabled = (balance > 0)
End Sub
```

How it works:

- The loop goes through each name in `symbols` and looks for a matching file, such as `Images\Cherry.png`.
- If a file is missing, a message tells you which one, and that reel shows the name as text so the game still runs. You need real pictures to pass the "Slots with images" check.
- `UpdateBalance` shows the balance and disables `btnSpin` when the balance is 0.

Step 6: The Spin button

This is the main part of the program: it checks the balance, takes the bet, spins the reels with random numbers and pays the winnings. Add it below `UpdateBalance`.

```
Private Sub btnSpin_Click(sender As Object, e As EventArgs) Handles
btnSpin.Click
    ' Cannot spin when the balance is 0
    If balance <= 0 Then
```

```
        MessageBox.Show("Your balance is $0. You cannot spin anymore!", "Out
of money")
        Return
    End If

    ' Read and check the bet
    Dim bet As Integer
    If Not Integer.TryParse(txtBet.Text, bet) OrElse bet < 1 OrElse bet >
balance Then
        MessageBox.Show("Please enter a whole number bet between 1 and " &
balance & ".", "Invalid bet")
        txtBet.Focus()
        Return
    End If

    balance -= bet

    ' Use random numbers to pick a symbol for each of the 3 reels
    Dim reels(2) As Integer
    For i As Integer = 0 To 2
        reels(i) = rnd.Next(0, symbols.Length)
    Next

    picReel1.Image = symbolImages(reels(0))
    picReel2.Image = symbolImages(reels(1))
    picReel3.Image = symbolImages(reels(2))

    ' Work out the multiplier
    Dim multiplier As Integer = 0
    If reels(0) = reels(1) AndAlso reels(1) = reels(2) Then
        multiplier = multipliers(reels(0))
        lblResult.Text = "JACKPOT! Three " & symbols(reels(0)) & "s - x" &
multiplier
    ElseIf reels(0) = reels(1) OrElse reels(1) = reels(2) OrElse reels(0) =
reels(2) Then
        multiplier = PAIR_MULTIPLIER
        lblResult.Text = "A pair! x" & multiplier & " (bet returned)"
    Else
        lblResult.Text = "No match. Better luck next time."
    End If

    Dim winnings As Integer = bet * multiplier
    balance += winnings
    lblResult.Text &= Environment.NewLine & "Bet: $" & bet & "   Won: $" &
winnings
```

```
    ' Save this spin in the history array
    AddToHistory(symbols(reels(0)) & " | " & symbols(reels(1)) & " | " &
symbols(reels(2)) &
                 "   Bet $" & bet & "  x" & multiplier & "  Won $" & winnings
&
                 "  Bal $" & balance)

    UpdateBalance()

    If balance = 0 Then
        MessageBox.Show("You are out of money! No more spins allowed.", "Game
over")
    End If
End Sub
```

How it works, in order:

1. **Balance check:** if the balance is 0, the spin stops with a message.
2. **Bet check:** `Integer.TryParse` makes sure the bet is a whole number from 1 up to the balance.
3. **Take the bet:** `balance -= bet`.
4. **Random reels:** `rnd.Next(0, symbols.Length)` returns a random index from 0 to 4 for each reel. That index picks the picture from `symbolImages`.
5. **Multiplier:** all three the same → that symbol's multiplier; any two the same → ×1; otherwise ×0.
6. **Winnings:** `winnings = bet * multiplier`, added to the balance.
7. **History and display:** the spin is saved with `AddToHistory` (Step 7) and the balance label is updated.

Example: you bet $10 and get Star | Star | Star. The bet is taken ($100 → $90), winnings are 10 × 10 = $100, and the new balance is $190.

Step 7: The spin history array

Every spin is stored in the `history` array, which holds a maximum of 10 entries. When it is full, every entry moves down one place so the oldest spin drops off and the newest goes in the last slot. Add these two subs below `btnSpin_Click`.

```
Private Sub AddToHistory(entry As String)
    If historyCount < MAX_HISTORY Then
        ' There is still room in the array
        history(historyCount) = entry
        historyCount += 1
```

```
    Else
        ' Array is full: shift everything left to drop the oldest spin
        For i As Integer = 0 To MAX_HISTORY - 2
            history(i) = history(i + 1)
        Next
        history(MAX_HISTORY - 1) = entry
    End If

    ShowHistory()
End Sub

Private Sub ShowHistory()
    ' Show the history array in the ListBox, newest spin at the top
    lstHistory.Items.Clear()
    For i As Integer = historyCount - 1 To 0 Step -1
        lstHistory.Items.Add(history(i))
    Next
End Sub
```

How the array changes on the 11th spin:

Index	Before (full)	After spin 11
0	Spin 1	Spin 2
1	Spin 2	Spin 3
…	…	…
8	Spin 9	Spin 10
9	Spin 10	Spin 11

`ShowHistory` copies the array into `lstHistory`, looping backwards so the newest spin is at the top. Your code is now complete: press **F5** to run it.

Step 8: Test before you submit

Run your program with **F5** and tick each box. These are the same checks your teacher will do.

Slots with images

- [ ] No "Missing pictures" message appears when the program starts.
- [ ] After a spin, all three reels show pictures.
- [ ] Spinning 5 times gives different results.

Balance

- [ ] The balance starts at $100.
- [ ] After a losing spin, the balance goes down by the bet.
- [ ] A bet of 0, a negative number, text, or more than the balance shows an "Invalid bet" message.
- [ ] Bet the whole balance until you lose it: at $0 a message appears and the SPIN button is disabled.

Winnings

- [ ] A pair returns the bet (balance stays the same as before the spin).
- [ ] Three of a kind pays bet × the multiplier from the payout table.
- [ ] The result label shows the bet and the amount won.

History

- [ ] Each spin adds one line to the history list, newest at the top.
- [ ] After 12 spins the list still has exactly 10 lines, and the first two spins are gone.

Troubleshooting

Problem	Fix
"Missing pictures" message, reels show words	Check the file names (`Cherry.png` , not `cherry.PNG` or `Cherry.png.png`), and that **Copy to Output Directory** is `Copy if newer` (Step 3).
`picReel1 is not declared` (or another control name)	The control's **(Name)** in the Properties window does not match the code. Rename it exactly as in Step 2.
Clicking SPIN does nothing	Check that the Sub ends with `Handles btnSpin.Click` , and that the button's name is `btnSpin` .
Two `btnSpin_Click` or `Form1_Load` subs	You double-clicked the control, which made an empty one. Delete the empty sub.
History shows more than 10 spins	Check `history(MAX_HISTORY - 1)` and the `Else` part of `AddToHistory` (Step 7).
Red error on `Image` or `Bitmap`	Make sure you created a **Windows Forms App (Visual Basic)**, not a Console App.